

From Theorem 10, for each vector b in $\mathbb{R}^m$, the matrix equation $Ax=b$ has at least one solution.

From Theorem 4, for each b in $\mathbb{R}^m$, the matrix equation $Ax=b$ has a solution if and only if the columns of A span $\mathbb{R}^m$.

$\therefore T$ maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$ $\square$

b) By Theorem 11, we know that T is one-to-one if and only if the equation $T(x)=0$ has only the trivial solution. By Theorem 10, we know the equation $T(x)=0$ has only the trivial solution if and only if $Ax=0$ has only the trivial solution. By defn. of matrix equation, if $A = [a_1 \ a_2 \ \dots \ a_n]$, $x_1 a_1 + x_2 a_2 + \dots + x_n a_n = 0$ if and only if $x_1 = x_2 = \dots = x_n = 0$.

By defn, the columns of A are linearly independent.

$\therefore T$ is one-to-one if and only if the columns of A are linearly independent $\square$

Statement (a) in Theorem 12 is equivalent to the statement " T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if every vector in $\mathbb{R}^m$ is a linear combination of the columns of A ."

Notation: Column vectors are written in rows, such as $x = (x_1, x_2)$, and $T(x)$ is written as $T(x_1, x_2)$.

Example 5: Let $T(x_1, x_2) = (3x_1 + x_2, 5x_1 + 7x_2, x_1 + 3x_2)$. Show ^{that} T is a one-to-one linear transformation. Does T map $\mathbb{R}^2$ onto $\mathbb{R}^3$?

Solution: $T(x) = \begin{bmatrix} 3x_1 + x_2 \\ 5x_1 + 7x_2 \\ x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} 3x_1 \\ 5x_1 \\ x_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ 7x_2 \\ 3x_2 \end{bmatrix}$ (Vector addition)

$$= x_1 \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix} \quad (\text{Scalar multiplication}) = \begin{bmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (\text{By defn. of } Ax)$$

By Theorem 10, $A = \begin{bmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{bmatrix}$ is the standard matrix for T .

We can easily prove that the columns of A are not multiples of each other and hence they are linearly independent.